A. Subset Mex

time limit per test 1 second
memory limit per test 512 megabytes
input standard input
output standard output

Rating: 900 | Tags: greedy, implementation, math | https://codeforces.com/problemset/problem/1406/A

Given a set of integers (it can contain equal elements).

You have to split it into two subsets A and B (both of them can contain equal elements or be empty). You have to maximize the value of $mex(A)+mex(B)$.

Here *mex* of a set denotes the smallest non-negative integer that doesn't exist in the set. For example:

- $mex(\{1,4,0,2,2,1\})=3$
- $mex(\{3,3,2,1,3,0,0\})=4$
- $mex(\text{varnothing})=0$ (*mex* for empty set)

The set is splitted into two subsets A and B if for any integer number x the number of occurrences of x into this set is equal to the sum of the number of occurrences of x into A and the number of occurrences of x into B.

Input

The input consists of multiple test cases. The first line contains an integer t ($1\le t\le 100$) — the number of test cases. The description of the test cases follows.

The first line of each test case contains an integer n ($1\le n\le 100$) — the size of the set.

The second line of each testcase contains n integers $a_1,a_2,\dots a_n$ ($0\le a_i\le 100$) — the numbers in the set.

Output

For each test case, print the maximum value of $mex(A)+mex(B)$.

Example

Input
4 6 0 2 1 5 0 1 3 0 1 2 4 0 2 0 1 6 1 2 3 4 5 6

Output
5 3 4 0

Note

In the first test case, $A=\{0,1,2\},B=\{0,1,5\}$ is a possible choice.

In the second test case, $A=\{0,1,2\},B=$varnothing is a possible choice.

In the third test case, $A=\{0,1,2\},B=\{0\}$ is a possible choice.

In the fourth test case, $A=\{1,3,5\},B=\{2,4,6\}$ is a possible choice.